



Dette er en gruppeopgave. Du skal først oprette en gruppe.

Dit svar IKKE AFLEVERET

Afleveringsopgave

The Second Mandatory Assignment constitutes the Exam Project.

The Exam Project will be developed in **groups of 4 or 5 students**. Please write down your names at [Groups – Exam Project.xlsx](#). If there are major circumstances that prevent you from working in groups, request a dispensation from Head of Education Marc Kluge (klu@ek.dk).

Each group of students will use the programming language(s), framework(s), libraries, database(s), and testing tool(s) of their choice to develop 5 independent APIs with the following requirements:

- The business scenario for each API will be chosen by the students
- All APIs must interact with database-stored information. Use the database model(s) and management system(s) of your choice
- All APIs must prevent SQL-injection, XSS, and CSRF
- APIs to develop:
 - REST
 - Endpoints. At least:
 - 3 GET requests that retrieve several items
 - They must implement search and pagination
 - 3 GET requests that retrieve one item by ID
 - 1 POST request
 - 1 PUT request
 - 1 DELETE request
 - A login endpoint
 - A logout endpoint
 - All mandatory RESTful constraints must be satisfied, including HATEOAS, which will provide contextual links
 - The REST naming convention will be followed
 - The API will be versioned. Use the format of your choice

- Authentication will be provided via OAuth2 with JWT
 - Choose a revocation strategy and implement it using a key-value store
- OpenAPI documentation will be implemented
- A test suite in Postman or a similar tool will be provided
 - It will include positive and negative requests
- SOAP
 - Operations. At least:
 - 2 that read data
 - 2 that change data
 - WSDL generation
 - Inclusion of at least 2 faults for error handling
 - A test suite in Postman or a similar tool will be provided
 - It will include positive and negative tests
- GraphQL
 - At least:
 - 2 queries
 - 2 mutations
 - SDL schema
 - A test suite in Postman or a similar tool will be provided
 - It will include positive and negative tests
- gRPC
 - At least:
 - 1 unary RPC
 - 1 bidirectional streaming RPC
 - Protobuf generation
 - A test suite in Postman or a similar tool will be provided
 - It will include positive and negative tests

- WebSockets
 - The same unary and bidirectional streaming operations as in gRPC
 - A test suite in Postman or a similar tool will be provided
 - It will include positive and negative tests

Deliverables

The source code of all 5 APIs, including the tests (collections or similar).

If the testing tool chosen does not allow to export collections for some type of API (e.g., gRPC in Postman), include screenshots in a pdf document.

Project Delivery

The final project will be structured in folders, one for each API. The project will be delivered to this Itslearning assignment [in a single zip file](#) on a group basis.

Do not include links (e.g., to Github repositories).

Deadline: **TBD, 23:59**